

Practical No. 3

Aim: To study and implement DML and DQL commands.

Software requirement: MySQL, Oracle 10g

Theory:

DML - Data Manipulation Language

The SQL commands that deal with the manipulation of data present in the database belong to DML or Data Manipulation Language and this includes most of the SQL statements. It is the component of the SQL statement that controls access to data and to the database.

1. INSERT Command:

INSERT command is used to insert data into a table.

Syntax: `INSERT INTO table_name VALUES (value1, value2, ...);`

Example: `INSERT INTO Students VALUES(01, 'Elon Musk', 'CSE');`

2. UPDATE Command:

UPDATE command is used to update existing data within a table.

Syntax: `UPDATE table_name SET column1 = value1, column2 = value2 WHERE condition;`

Example: `UPDATE Student SET name = 'Sam Altman' WHERE ID = 01;`

3. DELETE Command:

DELETE command is used to delete records from a database table.

Syntax: `DELETE FROM table_name WHERE condition;`

Example: `DELETE FROM Student WHERE ID = 01;`

DQL - Data Query Language

DQL statements are used for performing queries on the data within schema objects. The purpose of the DQL Command is to get some schema relation based on the query passed to it. This command allows getting the data out of the database to perform operations with it.

- **SELECT Command:**

SELECT command is used to retrieve data from the database.

Syntax: `SELECT column1, column2, ...FROM table_name WHERE condition;`

Example: `SELECT * FROM Student;`

`SELECT * FROM Student WHERE ID = 02;`

Program & Output:

```
CREATE TABLE student (ID INT, Name VARCHAR(50), Branch VARCHAR(50));
```

```
INSERT INTO student (ID, Name, Branch) VALUES (1, 'Elon Musk', 'CSE');
```

```
INSERT INTO student (ID, Name, Branch) VALUES (2, 'Sam Altman', 'AI');
```

```
INSERT INTO student (ID, Name, Branch) VALUES (3, 'Mark Zuckerberg', 'CSE(DS));
```

```
INSERT INTO student (ID, Name, Branch) VALUES (4, 'Jensen Huang', 'AIML');
```

```
SELECT * FROM student;
```

```
UPDATE student SET Name = 'Jeff Bezos' WHERE ID = 2;
```

```
SELECT * FROM student;
```

```
DELETE FROM student WHERE ID = 4;
```

```
SELECT * FROM student;
```

Output:

```
mysql> CREATE TABLE student (ID INT, Name VARCHAR(50), Branch VARCHAR(50));
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO student (ID, Name, Branch) VALUES (1, 'Elon Musk', 'CSE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student (ID, Name, Branch) VALUES (2, 'Sam Altman', 'AI');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student (ID, Name, Branch) VALUES (3, 'Mark Zuckerberg', 'CSE(DS)');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student (ID, Name, Branch) VALUES (4, 'Jensen Huang', 'AIML');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM student;
+-----+-----+-----+
| ID | Name          | Branch |
+-----+-----+-----+
| 1 | Elon Musk     | CSE    |
| 2 | Sam Altman    | AI     |
| 3 | Mark Zuckerberg | CSE(DS) |
| 4 | Jensen Huang  | AIML   |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> UPDATE student SET Name = 'Jeff Bezos' WHERE ID = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM student;
+-----+-----+-----+
| ID | Name          | Branch |
+-----+-----+-----+
| 1 | Elon Musk     | CSE    |
| 2 | Jeff Bezos    | AI     |
| 3 | Mark Zuckerberg | CSE(DS) |
| 4 | Jensen Huang  | AIML   |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> DELETE FROM student WHERE ID = 4;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM student;
+-----+-----+-----+
| ID | Name          | Branch |
+-----+-----+-----+
| 1 | Elon Musk     | CSE    |
| 2 | Jeff Bezos    | AI     |
| 3 | Mark Zuckerberg | CSE(DS) |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Conclusion:

Hence, we have successfully studied and implemented DML and DQL commands.